Strategic Blueprint for a Mature DevOps Ecosystem: Goals, KPIs, and Cross-Functional Alignment

Executive Summary: From Operational Maturity to Strategic Advantage

This report provides a comprehensive framework for the DevOps Chapter, which manages a large-scale, mature CI/CD ecosystem. The organization's current system, characterized by a high volume of over 10,000 daily builds and a 7 to 8-year history, is a testament to its foundational strength. The existing toolchain, including Jira for work management, Bitbucket for SCM, Jenkins with Shared Libraries, SonarQube for quality analysis, Artifactory for binary management, XLR for release orchestration, and Ansible, Docker, Terraform, and Datical for infrastructure, is robust and well-defined. The "isight" database, which has meticulously captured all CI/CD data for years, represents a critical and highly valuable asset.

The strategic imperative for the DevOps Chapter this year is to transcend its operational role and become a primary driver of business value. The goals for the year are not merely technical but are designed to directly align with key business outcomes. The proposed solution involves a strategic pivot from a reactive, tool-centric approach to a proactive, data-driven methodology. This requires formalizing the Chapter's mission, adopting a comprehensive framework of industry-standard metrics, addressing the challenges inherent in a long-standing system, and proactively bridging the gap between technical teams and the Line of Business (LOB).

This document serves as a strategic blueprint, outlining a phased roadmap to achieve these goals. It details a tiered metrics plan centered on DORA and DevSecOps, a diagnostic of critical challenges such as technical debt and pipeline sprawl, and a blueprint for fostering cross-functional excellence. The ultimate goal is to transform the "isight" data from a mere repository of logs into a source of predictive intelligence, thereby justifying technical investments and solidifying the DevOps Chapter's role as a strategic partner in the

organization's success.

Section 1: Defining a Strategic Vision for the DevOps Chapter

1.1 Core Mission and Foundational Goals

To establish a clear strategic direction for the year, the DevOps Chapter must first articulate a formal mission that moves beyond day-to-day operations. The core purpose of the Chapter is to enable the rapid, reliable, and secure delivery of software, which in turn accelerates innovation and provides direct value to the business. This mission serves as an evolution from a traditional, reactive role to a proactive, strategic one.[1]

This mission can be distilled into four foundational strategic pillars that will guide all activities and initiatives for the year:

- **Goal 1: Accelerate Time-to-Market:** This objective focuses on increasing the velocity of software delivery. By reducing the time it takes for new features or bug fixes to reach production, the organization can respond more quickly to customer needs and changing market conditions. This aligns with the Agile principles of continuous delivery and lean development.[1]
- **Goal 2: Enhance Operational Stability and Resilience:** A primary goal is to ensure the reliability of all systems and services. This involves maintaining high uptime, minimizing service disruptions, and ensuring a consistently positive user experience.
- **Goal 3: Proactively Manage Security and Compliance:** Security is a critical concern, especially in a large, mature system. This goal aims to embed security practices and controls throughout the development lifecycle, shifting from a reactive, late-stage concern to an integrated, "shift-left" discipline.[3]
- **Goal 4: Foster Cross-Functional Excellence:** A mature DevOps culture requires breaking down organizational silos. This pillar is dedicated to improving communication and collaboration across teams, particularly among DevOps Engineering, Site Reliability Engineering (SRE), Quality Assurance (QA), and the Line of Business (LOB). This cultivates a shared sense of ownership and responsibility for outcomes.[1]

These four goals are not isolated; they are interconnected and foundational to a high-performing software delivery organization. They are derived from the core principles of

DevOps, which emphasize automation, collaboration, and a holistic approach to the software development lifecycle.[7]

1.2 Leveraging the "isight" Data Foundation: From Logging to Predictive Analytics

The "isight" database, with its 7 to 8 years of historical data from over 10,000 daily builds, is the single most important asset the DevOps Chapter possesses. This extensive data set provides an unparalleled opportunity to move beyond basic monitoring and into advanced, predictive analytics.[9]

The current use of this data is likely reactive—diagnosing why a specific build failed or a deployment was slow. The new strategic goal is to transform this data repository into a source of proactive, predictive intelligence. This shift is enabled by the sheer volume and duration of the data, which allows for the application of advanced machine learning techniques.

A major strategic initiative for the DevOps Chapter this year should be to become a leader in the use of data analytics to optimize the CI/CD pipeline. With 10,000 builds per day, even a minor, recurring issue can represent a massive operational bottleneck. A human can't detect these subtle patterns, but a trained model can. The historical data from the "isight" database contains the patterns necessary to predict future events and proactively manage risks.

This strategic evolution can be broken down into three specific applications:

- **Predictive Failure Analysis:** Machine learning models can be trained on past build logs, test results, and commit histories to identify high-risk code changes or pipeline configurations before they are executed. For example, a model might analyze a specific combination of file changes and developer commit history to assign a "risk score" to a pull request. High-risk changes can then be flagged for additional scrutiny, while low-risk changes can be fast-tracked. This approach, which has been successfully implemented by companies like Netflix, can significantly reduce Change Failure Rate and prevent failures before they occur.[11]
- **Intelligent Test Selection:** For a large codebase with thousands of tests, running the full test suite on every commit is a major inefficiency that lengthens build times and increases lead time. By analyzing historical test results and code dependencies, a model can intelligently select and run only the most relevant tests for a given code change. This dramatically reduces test execution time without compromising bug detection accuracy, as demonstrated by Spotify's engineering teams.[12]
- **Resource Optimization:** The historical data can be used to forecast resource needs. By analyzing historical usage patterns, a model can predict peak times for builds or specific

projects that require more compute power. This allows for dynamic resource allocation and intelligent scheduling, leading to significant cost savings and improved throughput. Research has shown that such proactive, data-driven approaches can lead to a considerable reduction in infrastructure costs and build times.[11]

This strategic use of the "isight" data elevates the DevOps Chapter's role from a technical team to a key driver of organizational efficiency and innovation. It transforms a historical record into a powerful, forward-looking tool.

Section 2: A Comprehensive Framework of Metrics and KPIs

2.1 The Foundational DORA Framework: A Common Language for Velocity and Stability

To effectively measure progress and communicate value to the entire organization, the DevOps Chapter must adopt a standardized set of metrics. The DORA (DevOps Research and Assessment) framework provides an ideal starting point. DORA metrics are an evidence-based set of key measurements that have been shown to correlate directly with software delivery performance and overall organizational success.[15] Adopting this framework provides a common language that can be understood by technical teams and business stakeholders alike.

The DORA framework is composed of four key metrics, which measure two critical aspects of software delivery: velocity and stability.[8]

Velocity Metrics:

- **Deployment Frequency (DF):** This metric measures how often an organization successfully releases code to production. High-performing teams can deploy multiple times a day, while low-performing teams may deploy weekly or even less frequently.[8] A high deployment frequency indicates the team is delivering smaller, more manageable

batches of work, which reduces risk and enables faster feedback from users.[16]

- **Lead Time for Changes (LT):** This measures the average time it takes for a code change to go from committed to running successfully in production. This metric covers the entire value stream, including code review, testing, and deployment processes. A shorter lead time indicates an efficient development pipeline and the ability to respond quickly to market demands.[19] Elite performers can achieve a lead time of less than one hour, while low performers may take weeks or months.[8]

Stability Metrics:

- **Change Failure Rate (CFR):** This metric represents the percentage of deployments that result in a degraded service, requiring a hotfix, rollback, or other remediation. It is a key measure of the stability and quality of a team's software delivery process. A high CFR suggests potential issues in code quality, testing practices, or deployment procedures.[8] High-performing teams typically maintain a CFR of 0-15%.[8]
- **Mean Time to Restore (MTTR):** This metric measures the average time it takes to recover from a production failure or service interruption. A low MTTR demonstrates the team's efficiency in diagnosing problems, deploying fixes, and restoring service. High-performing teams can restore service in less than one hour.[8]

By establishing a baseline for these four metrics, the DevOps Chapter can objectively measure its current performance, identify key areas for improvement, and set clear, quantifiable goals for the year. The following table provides industry benchmarks and a placeholder for the organization's current baseline, which should be extracted from the "isight" database.

Metric	Definition	Elite Benchmark	High Benchmark	Medium Benchmark	Low Benchmark	Current isight Baseline
Deployment Frequency (DF)	How often code is successfully released to productio	Multiple deploys/day	1 per week to 1 per month	1 per month to 1 every 6 months	< 1 every 6 months	*[Placeholder for team data]*

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Lead Time for Changes (LT)	Time from code commit to code running in production	< 1 hour	1 day to 1 week	1 month to 6 months	> 6 months	*[Placeholder for team data]*
Change Failure Rate (CFR)	Percentage of deployments that cause a production failure	0-15%	16-30%	16-30%	> 30%	*[Placeholder for team data]*
Mean Time to Restore (MTTR)	Time to recover from a production service failure	< 1 hour	< 1 day	1 day to 1 week	> 6 months	*[Placeholder for team data]*

2.2 Evolving to DevSecOps Metrics: Embedding Security at Scale

In a large and mature system, security cannot be a reactive afterthought. It must be an integrated, proactive discipline. The practice of DevSecOps, which shifts security "left" into the development pipeline, is critical for managing security vulnerabilities and protecting the business from risk.[3] While DORA metrics provide a high-level view of stability, additional metrics are needed to quantify the effectiveness of security efforts.

These metrics go beyond standard security checks by quantifying the cost and risk of unaddressed vulnerabilities:

- **Mean Time to Remediate (MTTR-S):** This metric, distinct from the DORA MTTR, measures the average time it takes to fix a discovered security vulnerability. A short MTTR-S indicates an efficient and responsive security process.[21]
- **Mean Vulnerability Age (MVA):** MVA is the average age of all unresolved security vulnerabilities, from the time they are detected to the current moment. A low MVA is a strong indicator of a proactive security culture.[22]
- **Security Technical Debt (STD):** This metric represents the total number of unresolved vulnerabilities in production. It provides a clear, quantitative measure of the organization's ongoing security burden.[21]

The concept of Security Technical Debt (STD) is a powerful mechanism for aligning technical efforts with the Line of Business. It translates the abstract problem of "legacy code" or "unpatched systems" into a measurable, ongoing financial and reputational risk. A growing STD is a clear signal that the organization is accruing a dangerous, compounding interest, which can lead to significant data breaches, compliance penalties, and loss of customer trust. By tracking and actively working to reduce STD, the DevOps Chapter can demonstrate a direct contribution to the business's bottom line and brand reputation.

The organization's use of SonarQube is central to this effort. SonarQube's Quality Gates provide a critical, automated control point in the CI/CD pipeline.[23] These gates can be configured to fail a build if it introduces new security vulnerabilities, high-severity bugs, or code smells. A recommended best practice is to set a gate that, at a minimum, blocks any build with newly introduced critical or high-severity vulnerabilities. This "Clean as You Code" philosophy is essential for a large team, as it prevents the accumulation of new technical debt from the start.[23]

2.3 Toolchain-Specific Operational Metrics: A Deeper Dive

While DORA and DevSecOps metrics are essential for high-level strategy, a deeper layer of granular, tool-specific metrics is required for day-to-day operational management and targeted improvements. These metrics act as the levers that the DevOps Chapter can pull to improve the higher-level KPIs.

- **Jenkins (The Engine):** As the core automation engine, Jenkins' performance is paramount. Key metrics to track include: build duration, pipeline success/failure rate, queue time, and resource utilization (CPU/memory) of Jenkins agents.[11] Given the high volume of daily builds, efficiency at this level directly impacts Lead Time for Changes and overall throughput.
- **SonarQube (The Quality & Security Gate):** The metrics from SonarQube provide direct, actionable feedback to developers. These include: lines to cover, code coverage

on new code, number of code smells, and the count of bugs and vulnerabilities.[29] Monitoring these metrics ensures that quality and security are not compromised as the team accelerates its delivery.[23]

- **Artifactory (The Repository):** Artifactory's metrics are crucial for cost optimization and capacity planning. Tracking storage usage, number of components, and network egress provides visibility into resource consumption. For example, a single docker pull command can result in a dozen or more HTTP requests to Artifactory, which impacts network egress and overall performance.[31]
- **Jira & Bitbucket (The Workflow):** While these are not CI/CD tools, they contain valuable data that affects the DORA metrics. Tracking Pull Request size (number of code changes per request) and merge frequency can identify bottlenecks in the code review process that extend the Lead Time for Changes.[20]

The following table demonstrates how these granular, operational metrics directly map to the overarching strategic KPIs. This provides a blueprint for turning raw "isight" data into actionable intelligence. For example, a high number of "blocked threads" in Jenkins (an operational metric) can be traced to an increased "Mean Lead Time for Changes" (a strategic KPI), providing a clear path for investigation and resolution.

High-Level Strategic KPI	Low-Level Operational Metric	Tool	Causal Relationship
Deployment Frequency	Successful deployment count	Jenkins, XLR	An increase in successful deployments directly raises DF.
Lead Time for Changes	Pipeline duration, PR merge time	Jenkins, Bitbucket	Reducing build/test time and accelerating code review shortens LT.
Change Failure Rate	Build success rate, test failure rate	Jenkins, SonarQube	Failures in these stages prevent problematic changes from reaching production, thus lowering CFR.

Mean Time to Restore	Latency, Error rates (HTTP/400, 500)	System Monitoring	Fast detection of production errors allows for a quicker response, reducing MTTR.
Security Technical Debt	Unresolved vulnerabilities in production	SonarQube	Tracking and prioritizing these issues provides a clear path to reducing STD.

Section 3: Diagnosing and Mitigating Challenges in a Mature Ecosystem

A large-scale, long-standing CI/CD system, while powerful, is not without its challenges. The 7 to 8-year history of the organization's system means it likely suffers from issues that are a direct result of its age and scale. The DevOps Chapter must proactively diagnose and mitigate these challenges to ensure its strategic goals are met.

3.1 Managing Technical Debt and Legacy Systems

Technical debt is an inevitable consequence of any software project, and it becomes a primary concern in a system of this maturity. It is the result of prioritizing speed over quality, a lack of consistent standards, or a failure to modernize as technology evolves.[33] In a CI/CD pipeline, this debt manifests as: long lead times due to manual testing, inconsistent build environments, and a high Change Failure Rate. This can also lead to a cultural resistance to change, where teams prefer to stick to known, albeit inefficient, methods out of fear of breaking a fragile system.[34]

To address this, the Chapter must implement a structured approach to technical debt reduction:

- **Measure and Prioritize:** The first step is to quantify the debt. Metrics such as low code coverage, a high Change Failure Rate, or a large number of bugs reported by SonarQube

can be used to identify areas of highest impact.[33] A prioritization matrix can then be used to rank debt items based on severity and impact.[36]

- **Allocate Dedicated Time:** Just as financial debt is paid off with scheduled payments, technical debt must be addressed with dedicated time. A practice of allocating a percentage of each sprint to refactoring and clean-up tasks can prevent the debt from accumulating and ensures a continuous effort towards a healthier codebase.[33]
- **Implement "Shift Left" Controls:** The most effective way to manage technical debt is to prevent new debt from being introduced. This is where SonarQube's Quality Gates play a crucial role. By enforcing strict quality and security standards on all new code, the organization can reduce the likelihood of issues making it to production and can ensure that the codebase becomes cleaner over time.[3]

3.2 Combating CI/CD Pipeline Sprawl and Complexity

With over 10,000 builds a day, the system has likely experienced CI/CD pipeline "sprawl." This is a state where a proliferation of fragmented, inconsistent, and often manually-maintained pipelines creates inefficiency and risk. The user's mention of Jenkins Shared Libraries indicates an existing strategy to address this problem. Shared Libraries are designed to solve the "Don't Repeat Yourself" (DRY) problem by centralizing common build logic.[38]

However, there are risks associated with this approach. Without proper management, a Shared Library can become a large, monolithic, and difficult-to-test codebase. A single bug or performance issue in the library can break hundreds or thousands of pipelines simultaneously. This presents a unique challenge that must be addressed by professionalizing the Shared Library's development lifecycle. The library itself should be treated not as a simple collection of scripts, but as a mission-critical, self-contained application.

The following practices should be adopted for the Shared Library:

- **Unit Testing:** Implement a unit testing framework for the Shared Library's Groovy classes. This allows developers to test code locally before pushing changes, preventing bugs from affecting the entire CI/CD ecosystem.[40]
- **Version Control:** Utilize Git tags or branches to version the library. This allows pipelines to pin to a specific, stable version of the library, providing control over rollouts and enabling quick rollbacks if an issue is discovered.[38]
- **Performance Optimization:** As a central component, the Shared Library's performance is critical. Complex Groovy code should be avoided in favor of invoking external scripts via the sh or bat steps, which offloads the work from the Jenkins controller to the build agent.[14] This reduces resource consumption on the controller and improves overall

pipeline speed.

By adopting these practices, the Chapter can successfully combat pipeline sprawl, ensure consistency, and prevent the Shared Library from becoming a new source of technical debt.

3.3 Overcoming Organizational Silos and Resistance to Change

Beyond the technical challenges, the most significant hurdles in a mature organization are often cultural. Siloed teams, a lack of communication, and a fear of change can hinder even the best technical initiatives.[6] The DevOps Chapter's success hinges on its ability to build bridges between departments and create a culture of shared responsibility.

- **Establish Clear Communication Channels:** Effective collaboration requires clear, consistent communication. Utilizing a central work management tool like Jira and a communication platform like Slack can facilitate transparent conversations and break down the "siloed communication" anti-pattern.[2]
- **Foster Shared Responsibility:** The "you build it, you run it" philosophy is a core tenet of DevOps. This culture encourages development teams to take ownership of their code from commit to production, reducing the reliance on a single Operations team and fostering a greater sense of shared accountability for software quality and stability.[2]
- **Encourage Cross-Skilling:** To break down knowledge silos, the Chapter should encourage and invest in cross-skilling. This means providing opportunities for engineers to learn about other domains—for example, for a developer to gain a better understanding of SRE practices or for a QA engineer to learn about infrastructure automation. This builds empathy and improves collaboration by creating a more holistic understanding of the end-to-end delivery process.[2]

Section 4: A Blueprint for Cross-Functional Alignment

The ultimate success of the DevOps Chapter's goals depends on its ability to align the efforts of diverse technical teams and demonstrate its value to the business. This requires a shared understanding of objectives and a common language for measuring success.

4.1 Bridging the Gap: DevOps Engineering, SRE, and QA

In many organizations, the relationship between these three teams can be a source of tension due to seemingly conflicting goals. The DevOps Engineering team is often focused on maximizing deployment frequency and speed, while the SRE team's primary mission is to maintain the stability and reliability of the production environment, using metrics like Service Level Objectives (SLOs).[7] This can create a friction point where speed is seen as a threat to reliability.

The key to bridging this gap lies in the adoption of shared metrics. The DORA framework provides the harmonizing mechanism that connects the goals of speed and stability. For example, a high Deployment Frequency coupled with a low Change Failure Rate proves that high velocity does not have to come at the expense of reliability. These metrics become the single source of truth for all three teams, fostering collaboration and demonstrating that they are working towards a common objective.

SRE's "Four Golden Signals"—Latency, Traffic, Errors, and Saturation—provide the low-level, real-time data needed to understand the root causes of issues that impact DORA metrics.[44] For example, if the Change Failure Rate suddenly spikes, the SRE team can analyze the "Errors" signal to quickly identify the root cause, providing a direct, collaborative feedback loop to the development teams. This approach turns a potential conflict into a productive, data-driven partnership, enabling teams to work together to reduce Mean Time to Restore.

4.2 From Technical Metrics to Business Value: Aligning with the Line of Business (LOB)

The Line of Business is concerned with outcomes that directly impact revenue, customer satisfaction, and market share. They do not typically track build duration or number of daily builds. Therefore, the DevOps Chapter must proactively translate its technical achievements into a language that the business understands.[45]

The DORA and DevSecOps metrics are a powerful tool for this translation. Each technical KPI can be mapped to a tangible business benefit, providing a clear justification for investments and a way to demonstrate the value of DevOps initiatives.

The following table demonstrates this critical alignment:

Technical KPI	LOB-Facing Business Outcome

Shorter Lead Time for Changes	Enables the business to respond to customer feedback and market demands more quickly, leading to faster feature delivery and a competitive advantage.
Reduced Mean Time to Restore	This translates directly to reduced system downtime, improved customer experience, and minimized lost revenue due to outages.
Lower Security Technical Debt	Demonstrates a proactive reduction in security risk, protects customer data, and ensures compliance with regulations, safeguarding brand reputation.
Increased Deployment Frequency	Allows for the delivery of smaller, safer changes, which reduces risk and enables faster feedback loops from customers, accelerating innovation.

By consistently communicating and reporting on these metrics in business terms, the DevOps Chapter can shift its perception from a cost center to a strategic partner in the organization's success.[8]

Section 5: Strategic Roadmap and Action Plan

To achieve the ambitious goals outlined in this report, the DevOps Chapter should adopt a phased, crawl-walk-run approach. This strategy prevents the organization from being overwhelmed and builds momentum through a series of achievable, high-impact activities.

5.1 Proposed Roadmap for the Year

Quarter 1: Baseline and Assessment (Crawl)

- **Objective:** Establish a data-driven foundation and a common understanding of goals.
- **Activities:**
  - **Data Extraction:** Use the "isight" database to extract and analyze the current baselines for all four DORA metrics. This establishes a clear starting point for all future improvements.[8]
  - **Technical Audit:** Conduct a comprehensive audit of existing pipelines and the Jenkins Shared Library to identify technical debt, complexity, and areas for consolidation.[33]
  - **Cross-Functional Workshop:** Host a workshop to introduce the DORA framework and the strategic goals to DevOps Engineering, SRE, and QA teams. This ensures early alignment and a shared understanding of the year's mission.[2]

Quarter 2: Foundational Improvements (Walk)

- **Objective:** Address high-impact, low-effort bottlenecks and implement key controls.
- **Activities:**
  - **Quality Gate Implementation:** Implement SonarQube Quality Gates on all pipelines, focusing initially on failing builds that introduce new critical or high-severity issues.[23]
  - **Technical Debt Reduction:** Start a structured program to address the identified technical debt. Prioritize issues that have the highest impact on improving DORA metrics, such as refactoring a legacy build process that significantly contributes to a long Lead Time for Changes.[36]
  - **Pipeline Consolidation:** Begin the process of migrating fragmented and duplicate pipelines to the Jenkins Shared Library, which will improve consistency and centralize maintenance.[38]

Quarter 3: Advanced Analytics & Optimization (Run)

- **Objective:** Evolve to a proactive, intelligent CI/CD system.
- **Activities:**
  - **Predictive Analytics:** Begin to leverage the historical "isight" data by training machine learning models to predict pipeline failures and optimize resource allocation.[11] This initiative directly contributes to reducing Change Failure Rate and build duration.
  - **Professionalize Shared Library:** Implement a professional development lifecycle for

the Shared Library, including unit testing for core functions and a versioning strategy using Git tags.[38] This ensures the library's stability and prevents it from becoming a source of new technical debt.
    - **SRE Collaboration:** The DevOps Chapter should work closely with the SRE team to map DORA metrics to the "Four Golden Signals" to enable a clear, real-time feedback loop for incident response.[7]

Quarter 4: Cultural & Strategic Alignment (Scale)

- **Objective:** Cement the new culture and demonstrate business value.
- **Activities:**
    - **Unified Dashboards:** Create and disseminate cross-functional dashboards that display shared DORA and DevSecOps metrics. These dashboards should be accessible to all teams and provide a single source of truth for delivery performance.[8]
    - **LOB Reporting:** Prepare a year-end report for the LOB and senior leadership. This report should explicitly map the year's achievements in DORA, DevSecOps, and operational metrics to tangible business outcomes such as reduced operational costs, faster time-to-market, and improved customer satisfaction.[32]
    - **Future Planning:** Use the year's data and achievements to define the strategic goals for the next year, ensuring the momentum of continuous improvement is maintained.

5.2 Final Recommendations

Based on the analysis, the following actions are recommended for the DevOps Chapter:

- **Prioritize DORA as the North Star:** The four DORA metrics are the ideal starting point. They are simple to understand, correlate with business value, and provide a clear, objective measure of performance. All improvement efforts should be framed in terms of how they impact these core KPIs.
- **Treat "isight" as a Strategic Asset:** The "isight" database is the most valuable tool the Chapter has. Investing in turning this raw data into a source of predictive intelligence is a high-impact initiative that will differentiate the organization's DevOps capabilities.
- **Address the Paradox of Shared Libraries:** While a powerful tool, the Jenkins Shared Library must be treated as a professional, versioned, and tested application. This prevents a single point of failure from impacting the entire pipeline and ensures long-term scalability and maintainability.
- **Think in Business Terms:** All initiatives must be measured and reported not just in terms

of technical success, but in terms of the business value they create. Proactively translating metrics into outcomes that resonate with the Line of Business is crucial for securing continued investment and support.

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